Laurent Juliá Calac

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EDUCATION

Georgetown University *Washington, D.C.*
M.S. in Data Science & Analytics, Concentration in AI | GPA: 4.0 *Expected May 2027*
Relevant Coursework: Neural Networks & Deep Learning, Natural Language Processing, Reinforcement Learning, Big Data & Cloud Computing, Applied Generative AI

Georgia Southern University *Statesboro, Georgia*
B.S. in Information Technology | GPA: 3.75, Magna Cum Laude *Jan 2022 – Jul 2025*

TECHNICAL SKILLS

Programming & Data: Python (Pandas, Polars, NumPy, scikit-learn), SQL, R
Machine Learning & AI: PyTorch, TensorFlow, Hugging Face, LLMs, Ollama, LLM Evaluation, Agentic AI, CNNs, NLP, YOLO
Data Engineering & Analytics: SQL Server, SQLite, DuckDB, Parquet, Spark, AWS (EC2), ETL, Streamlit, Power BI, Git/GitHub, CI/CD
Languages: English, German, Spanish (bilingual proficiency); French (limited working proficiency)

WORK EXPERIENCE

Foerster Group *Reutlingen, Germany*
AI & Data Science Intern *Jun 2026 – Aug 2026*

- Architected an end-to-end ETL pipeline in Python, SQL Server, and SQLite that unified 20 years of historical inspection data (4,000+ files from 2,000+ inspections) from network-stored backups, Excel defect files, and PDF/Word reports into one platform.
- Automated SQL Server backup ingestion with a dynamic schema that auto-discovers tables and adapts to schema drift across software versions, eliminating manual database restoration.
- Built an Excel parsing engine handling multi-row headers and shifting layouts across 6 inspection types, converting inconsistent spreadsheets into structured, analysis-ready records.
- Developed a prototype Streamlit dashboard with cascading drill-down filters and automated KPI/status logic, consolidating each inspection's data from scattered file types into a single view for ~10 inspection engineers.
- Benchmarked general-purpose and text-to-SQL LLMs via Ollama and Hugging Face on execution accuracy, SQL validity, and latency for natural-language querying of the inspection database; top models reached 70% execution accuracy, and results guided the planned LLM integration.

Grupo Mallén *Santo Domingo, Dominican Republic*
Data Analytics Intern *Jul 2025 – Aug 2025*

- Led BI consolidation, reducing 20 Power BI dashboards to 5 executive-ready views for the CFO and senior management, cutting reporting complexity by 75%.
- Partnered with the CFO and marketing team on sales, marketing, and budgeting dashboards that accelerated decision-making.

Georgia Southern University *Statesboro, Georgia*
Data Analytics Intern *May 2025 – Jul 2025*

- Automated weekly IT help-ticket reporting with Python, SQL, Excel, and Power BI, tracking ticket volume, resolver, and resolution time.
- Created Power BI dashboards tracking invoice volume, cycle time, on-time rate, exceptions, and staffing for system leadership.

PROJECTS

ECG Arrhythmia Classification | CNNs & Transformers *Apr 2026*

- Developed and compared CNN, Transformer, and hybrid models for 4-class ECG arrhythmia classification, achieving 99.0% test accuracy and 0.93 Macro F1.
- Designed preprocessing pipelines for waveforms, spectrograms, and metadata that addressed class imbalance; improved Macro F1 from 0.71 to 0.86 using Transformers over CNNs on raw waveforms.

Gender Bias Analysis in Wikipedia Biographies | NLP & Fairness Research *Dec 2025*

- Analyzed 573K+ Wikipedia biographies with a modular NLP pipeline (scraping, preprocessing, inference, analysis) built with Poetry and pytest, using VADER, TextBlob, and RoBERTa.
- Identified statistically significant gender differences using t-tests, KS-tests, Mann–Whitney U, and χ^2 tests.

LEADERSHIP

Division 1 Student Athlete, Georgia Southern Men's Tennis Team, *Team Captain* *Jan 2022 – May 2025*

- Balanced a 20-hour weekly training schedule with full-time academics; captained the team to its first-ever Sun Belt Regular Season Championship.

Graduate Student Mentor, Georgetown University, M.S. Data Science & Analytics *Sep 2026 – Present*
Volunteer Graduate Assistant, Georgetown Men's & Women's Tennis Programs *Sep 2025 – Present*